

ADITYA PRABHAT GUPTA

Full Stack Engineer

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SUMMARY

Full Stack Engineer with hands-on experience building scalable real-time and AI-powered web applications using modern JavaScript technologies. Skilled in designing responsive frontend architectures, secure backend systems, REST APIs, and low-latency real-time communication using WebRTC and Socket.IO. Strong understanding of authentication, cloud deployment, performance optimization, and scalable application development.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3

Frontend: React.js, Vite, Tailwind CSS, React Router DOM, Axios, Framer Motion

Backend: Node.js, Express.js, REST APIs, Socket.IO, WebRTC, JWT, bcryptjs, Redis

Databases: MongoDB (Mongoose), MySQL

Cloud & Tools: Git, GitHub, Docker, Cloudinary, FFmpeg, Sharp, Vercel, Render, Netlify, Postman

AI & ML: TensorFlow, Prompt Engineering, AI Integration, Groq API, Gemini AI

EDUCATION

Roorkee Institute of Technology

B.Tech in Computer Science and Engineering (AI & ML)

Roorkee, Uttarakhand

Aug 2023 – July 2027

EXPERIENCE

Artificial Intelligence & Generative AI Intern

YBI Foundation

June 2025 – July 2025

Remote

- Engineered AI-powered applications using Generative AI APIs and prompt engineering workflows
- Implemented machine learning workflows involving model experimentation, debugging, and AI-based solution deployment

PROJECTS

Convex – Real-Time Chat & Calling App | MERN, Socket.IO, WebRTC, Redis, Cloudinary | [GitHub](#) | [Live](#)

Apr 2026

- Engineered a real-time messaging and calling platform using **WebRTC** and **Socket.IO**; load-tested to **50 stable concurrent WebSocket connections** with avg. **180ms message latency** via custom STUN/TURN configuration
- Integrated **Google OAuth 2.0** and **JWT Authentication** for secure user access and session management
- Benchmarked media pipeline (Sharp + FFmpeg + Cloudinary) across 20 sample files; achieved average **58% reduction in upload size**, cutting image processing time from **1.4s to under 0.6s**
- Integrated AI-powered conversational assistance using Groq API and implemented Redis-backed session caching, reducing repeated DB lookups by **40%**; deployed via Docker with environment-separated configs

Project Management System | React.js, Node.js, Express.js, MongoDB, JWT | [GitHub](#) | [Live](#)

Jan 2026

- Developed a MERN-based project management platform featuring a **Kanban Board**, task tracking, and Role-Based Access Control (**admin / member / viewer**) with per-route middleware enforcement
- Designed **12 RESTful API endpoints** for project, task, and user management; secured with JWT and validated via express-validator, keeping avg. response time **under 120ms** on Render free tier
- Implemented optimistic UI updates on the Kanban board, reducing perceived drag-and-drop latency and eliminating full-page re-renders on task state changes

RecipeX – Recipe Discovery Platform | React.js, React Router, TheMealDB API | [GitHub](#) | [Live](#)

Dec 2025

- Built a recipe discovery platform with category-based filtering and dynamic search functionality
- Implemented **400ms debounce** on search input, eliminating redundant API calls on partial keystrokes and reducing network requests during active typing sessions
- Added protected routes and persistent favorites via localStorage with graceful fallback for private browsing

AI Medical Assistant Chatbot | Python, TensorFlow, JavaScript, Groq API | [GitHub](#) | [Live](#)

Mar 2025

- Led a **3-member team** to develop an NLP-based healthcare chatbot using **TensorFlow** for intent classification; trained on **15 symptom categories** achieving **87% validation accuracy** on held-out test set
- Enhanced platform with Groq API for open-ended medical Q&A, health quiz modules, wellness recommendations, and secure **JWT Authentication**
- Secured **Runner-Up** in RIT Hackathon 2025 among **20+ participating teams**

ACHIEVEMENTS & CERTIFICATIONS

3rd Place – VibeCon '26 AI Creator Showcase (GDGoC RIT Roorkee) – Built DreamForge AI using Gemini AI, Imagen API, React.js, and Framer Motion

2nd Runner-Up – RIT Hackathon 2025 – Developed NLP-based AI Medical Assistant Chatbot using TensorFlow

Full Stack Development with AI – Physics Wallah x NSDC x PwC (2026) | [Credential Link](#)

Core Competencies: Full-Stack Development, REST API Design, Authentication & Security, Real-Time Systems